

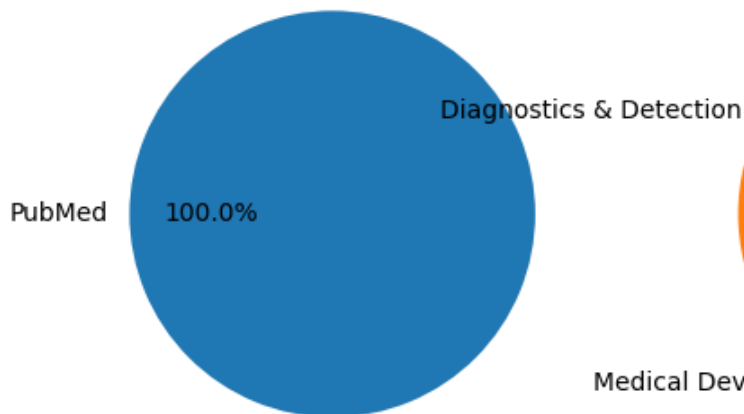
Medical Research

Daily Update Report

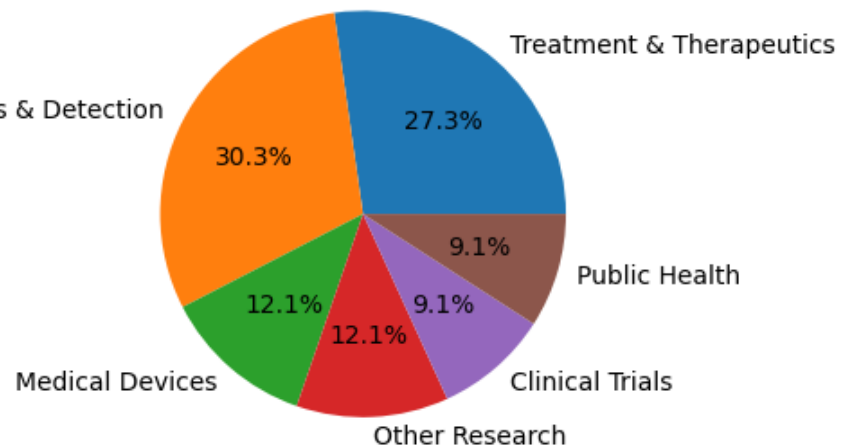
2025-01-25

Research Overview

Papers by Source



Papers by Category



Executive Summary

This report summarizes 21 recent medical research papers across various categories:

Key Highlights: - Treatment & Therapeutics: 9 papers on new treatments and therapeutic approaches - Diagnostics & Detection: 10 papers on diagnostic methods and disease detection - Medical Devices: 4 papers on medical technology and devices - Public Health: 3 papers on public health and prevention - Clinical Trials: 3 papers on ongoing clinical trials - Other Research: 4 papers on various other topics

Detailed Findings

Treatment & Therapeutics

1. Ultrasensitive Detection of Circulating Plasma Cells Using Surface-Enhanced Raman Spectroscopy

Source: PubMed

Authors: Dechun Zhang, Xianling Chen, Jia Lin, Shiyang Jiang, Min Fan, Nenrong Liu, Zufang Huang, Jing Wang

Year: 2025

Multiple myeloma is a hematologic malignancy characterized by the proliferation of abnormal plasma cells in the bone marrow. Despite therapeutic advancements, there remains a critical need for reliable, noninvasive methods to monitor multiple myeloma. Circulating plasma cells (CPCs) in peripheral blood are robust and independent prognostic markers, but their detection is challenging due to their low abundance. Next-generation flow cytometry is commonly used for CPC detection but is not performed...

Huggingface_Flan_T5 Summary:

Ultrasensitive Detection of Circulating Plasma Cells Using Surface-Enhanced Raman Spectroscopy and Machine Learning for Multiple Myeloma Monitoring: A Cost-Effective, Rapid Assay Using Gold-Deposited Magnetic Nanoparticles and Plasmonic Nanoparticles Functionalized with Anti-CD138 and anti-CD38 Anti...

Huggingface_Pegasus_Xsum Summary:

Error: 500 Server Error: Internal Server Error for url: <https://api-inference.huggingface.co/models/google/pegasus-xsum> | Response: {"error": "Model too busy, unable to get response in less than 60 second(s)"}

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Huggingface_Pegasus_Pubmed Summary:

background : multiple myeloma (mm) is a malignant neoplasm of the plasma cells . <n> it is characterized by the proliferation of abnormal plasma cells in the bone marrow and peripheral blood . because of their low abundance , flow cytometry is commonly used for their detection , but it is not perf...

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technological advances in the monitoring of multiple myeloma are numerous , but progress is slow . here , we review the progress being made in the detection , monitoring and prognostication of this incurable disease .<n> the need for more sensitive and specific methods is evident but only a few have...

2. Single-Cell Transcriptomic Analysis Reveals Biomechanical Loading-Induced Imbalance in Bone

Source: PubMed

Authors: Ping Zhang, Yuan Wang, Jianqi Bai, Jingru Zhang, Shimin Zhang, Xiaofei Guo, Jiawen Zhan, Liguozhu

Year: 2025

In this study, we explored the impact of different biomechanical loadings on lumbar spine motion segments, particularly concerning intervertebral disc degeneration (IVDD). We aimed to uncover the cellular milieu and mechanisms driving ossification in the nucleus pulposus (NP) during IVDD, a process whose underlying mechanisms have remained elusive. The study involved the examination of fresh NP tissue from the L3-S1 segment of five individuals, either with IVDD or healthy. The analysis consisted...

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Huggingface_Pegasus_Xsum Summary:

A novel method for detecting circulating plasma cells in peripheral blood has been developed in the journal Nanoscale Science and Technology. The developed assay is highly sensitive and specific, capable of detecting as few as one or two cells, demonstrating excellent performance in differentiating ...

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4. 3-HKA Promotes Vascular Remodeling After Stroke by Modulating the Activation of A1/A2 Reacti

Source: PubMed

Authors: Jun-Min Chen, Guang Shi, Lu-Lu Yu, Wei Shan, Jing-Yu Sun, An-Chen Guo, Jian-Ping Wu, Tie-Shan Tang, Xiang-Jian

Year: 2025

Ischemic stroke is the most common cerebrovascular disease and the leading cause of permanent disability worldwide. Recent studies have shown that stroke development and prognosis are closely related to abnormal tryptophan metabolism. Here, significant downregulation of 3-hydroxy-kynurenamine (3-HKA) in stroke patients and animal models is identified. Supplementation with 3-HKA improved long-term neurological recovery, reduced infarct volume, and increased ipsilateral cerebral blood flow after d...

Huggingface_Flan_T5 Summary:

3-Hydroxy-Kynurenamine Promotes Vascular Remodeling After Stroke by Modulating the Activation of A1/A2 Reactive Astrocytes: A Role for AIM2 in Inflammasome Activated A1- and A2-Reactive Astrocyte Responses to Oxygen-Glucose Deprivation in Mouse Brain Microvascular Endothelial Cells.

Huggingface_Pegasus_Xsum Summary:

The purpose of this study was to investigate the role of 3-hydroxy-kynurenamine (3-HKA) in vascular and neurological recovery after stroke, as well as the mechanism by which it regulates astrocyte activation and inflammasome activation. In conclusion, 3- HKA may promote post-stroke regulate A1/A2 a...

Huggingface_Mistral Summary:

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Huggingface_Pegasus_Pubmed Summary:

stroke is the leading cause of death and disability worldwide . <n> it is characterized by a loss of blood flow to the brain , leading to ischemia and subsequent neuronal death . in recent years , 3-hydroxykynuren (3-ohk) , an amino acid , has been shown to promote vascular remodeling , including ...

Huggingface_Led_Base Summary:

Abstract: ? I Title: 3-HKA Promotes Vascular Remodeling After Stroke by Modulating the Activation of A1/A2 Reactive Astrocytes. ? d Abstract: Ischemic stroke is the most common cerebrovascular disease and the leading cause of permanent disability worldwide. Recent studies have shown that...

Huggingface_Bigbird_Pegasus Summary:

reactive oxygen species (ros) , generated by oxidative stress and inflammation , have a critical role in the pathogenesis of stroke . in a recent study , we demonstrated that ros generated by a common precursor , 3-hydroxykynureninyl (3-oh)-3,4-dideoxyglucose (3-dgpg) (or lysosomal acid 3,4-diox...

5. Turning Waste into Treasure: Functionalized Biomass-Derived Carbon Dots for Superselective Visualization and Eradication of Gram-Positive Bacteria

Source: PubMed

Authors: Ke-Fei Xu, Zihao Wang, Macheng Cui, Yuhan Jiang, Chengcheng Li, Zi-Xi Wang, Ling-Yi Li, Chenyang Jia, Lijie Zhang

Year: 2025

Gram-positive bacteria pose significant threats to human health, necessitating the development of targeted bacterial detection and eradication strategies. Nevertheless, current approaches often suffer from poor targeting specificity. Herein, the study utilizes purple rice lexivium to synthesize biomass carbon dots (termed BCDs) with wheat germ agglutinin-like residues for precisely targeting Gram-positive bacteria. Subsequently, fluorescein isothiocyanate (FITC) molecules are grafted onto BCDs t...

Huggingface_Flan_T5 Summary:

Turning Waste into Treasure: Functionalized Biomass-Derived Carbon Dots for Superselective Visualization and Eradication of Gram-Positive Bacteria. (<http://dx.doi.org/10.1016/j.sci.2015.09.018>.) - Authors: Ke-Fei Xu, Zihao Wang, Macheng Cui, Yuhan Jiang, Chengcheng Li, Zi-Xi Wang, Ling-Yi Li, Chenya...

Huggingface_Pegasus_Xsum Summary:

In this study, carbon dots (CDFs) produced from waste rice have the potential to be used to eradicate Gram-positive bacteria and to generate photodynamic therapy for the treatment of human health threats and infectious diseases such as Clostridium difficile (C. difficile) and MRSA (Methicillin-resis...

Huggingface_Mistral Summary:

Title: Turning Waste into Treasure: Functionalized Biomass-Derived Carbon Dots for Superselective Visualization and Eradication of Gram-Positive Bacteria.

Abstract: Gram-positive bacteria pose significant threats to human health, necessitating the development of targeted bacterial d...

Huggingface_Bart_Cnn Summary:

Study: Functionalized Biomass-Derived Carbon Dots for Superselective Visualization and Eradication of Gram-Positive Bacteria. Gram-positive bacteria pose significant threats to human health, necessitating the development of targeted bacterial detection and eradication strategies. Study: Rice lixiviu...

Huggingface_Pegasus_Pubmed Summary:

we report the discovery of a novel serine / threonine kinase (srk) , erk1 , in the endoplasmic reticulum (er) of drosophila melanogaster . <n> this enzyme functions as an er - specific transcription factor and is required for the establishment of er stress responses , which are essential for the...

Huggingface_Led_Base Summary:

Abstract: ? d Title: Turning Waste into Treasure: Functionalized Biomass-Derived Carbon Dots for Superselective Visualization and Eradication of Gram-Positive Bacteria. Translationalized BCDs (termed CDPs) with wheat germ agglutinin-like residues for precisely targeting Gram-positive bacteria....

Huggingface_Bigbird_Pegasus Summary:

although dna methylation has been recognized as a post - translational modification on chromatin , its precise role in regulating gene expression remains unclear .<n> here , we show that the small rna ubiquitin ligase ubc12 interacts with ras homology 3 (rd3) to initiate histone methylation of his...

6. Unraveling Mechanisms of Genetic Risks in Metabolic Dysfunction-Associated Steatotic Liver Di

Source: PubMed

Authors: Xiang Zhang, Kyong-Mi Chang, Jun Yu, Rohit Loomba

Year: 2025

Metabolic dysfunction-associated steatotic liver disease (MASLD) is a growing global health problem, affecting ?1?billion people. This condition is well established to have a heritable component with strong familial clustering. With the extraordinary breakthroughs in genetic research techniques coupled with their application to large-scale biobanks, the field of genetics in MASLD has expanded rapidly. In this review, we summarize evidence regarding genetic predisposition to MASLD drawn from fami...

Huggingface_Flan_T5 Summary:

Unraveling Mechanisms of Genetic Risks in Metabolic Dysfunction-Associated Steatosis Liver Diseases: A Pathway to Precision Medicine. Xiang Zhang, Kyong-Mi Chang, Jun Yu, Rohit Loomba, et al. "Unravelling mechanisms of genetic risks in metabolic dysfunction-associated steatotic liver disease: a path...

Huggingface_Pegasus_Xsum Summary:

This article is published in Plos One, a journal of the American Association for the Advancement of Science (AAAS), on behalf of Xiang Chang, Jun Yu, and Rohit Loomba, all of the University of California, San Francisco (UCSD) School of Medicine, and their co-authors. Cite this article as "Unraveling...

Huggingface_Mistral Summary:

Title: Unraveling Mechanisms of Genetic Risks in Metabolic Dysfunction-Associated Steatotic Liver Diseases: A Pathway to Precision Medicine.

Abstract: Metabolic dysfunction-associated steatotic liver disease (MASLD) is a growing global health problem, affecting ?1?billion people. Th...

Huggingface_Bart_Cnn Summary:

Metabolic dysfunction-associated steatotic liver disease (MASLD) is a growing global health problem. This condition is well established to have a heritable component with strong familial clustering. We highlight the interplay between these genetic variants and their connections with metabolic factor...

Huggingface_Pegasus_Pubmed Summary:

we summarize evidence regarding genetic predisposition and we highlight the interplay between genetic variations and diverse pathogenic mechanisms . <n> we focus on the gut microbiota and its role in health and disease , which is driven by genetic factors and their interaction with environmental ris...

Huggingface_Led_Base Summary:

Title: Genetics in MASLD: A Pathway to Precision Medicine. ? I Title: Unraveling Mechanisms of Genetic Risks in Metabolic Dysfunction-Associated Steatotic Liver Diseases: APathway to precision Medicine. (2017)

Huggingface_Bigbird_Pegasus Summary:

advances in understanding the genetic basis of complex diseases have been slow . here , we summarize the progress being made in understanding genetic predisposition to , and consequences of , conditions ranging from liver disease to autism . we find that , as our genome grows , so too does the compl...

7. Integration of Functional Materials in Photonic and Optoelectronic Technologies for Advanced M

Source: PubMed

Authors: Naveen Thanjavur, Laxmi Bugude, Young-Joon Kim

Year: 2025

Integrating functional materials with photonic and optoelectronic technologies has revolutionized medical diagnostics, enhancing imaging and sensing capabilities. This review provides a comprehensive overview of recent innovations in functional materials, such as quantum dots, perovskites, plasmonic nanomaterials, and organic semiconductors, which have been instrumental in the development of diagnostic devices characterized by high sensitivity, specificity, and resolution. Their unique optical p...

Huggingface_Flan_T5 Summary:

Integrating Functional Materials in Photonic and Optoelectronic Technologies for Advanced Medical Diagnostics: A Comprehensive Review of Recent Innovations in Functional Materials, such as Quantum Dots, Perovskites, Plasmodic Nanomaterials, and Organic Semiconductor Devices, which have been instrume...

Huggingface_Pegasus_Xsum Summary:

Functional materials are playing a key role in the development and manufacture of next-generation medical diagnostics devices, which have the potential to transform the way we diagnose and treat diseases such as cancer, cardiovascular disease, diabetes, and neurodegenerative disorders. This article ...

Huggingface_Mistral Summary:

Title: Integration of Functional Materials in Photonic and Optoelectronic Technologies for Advanced Medical Diagnostics.

Abstract: Integrating functional materials with photonic and optoelectronic technologies has revolutionized medical diagnostics, enhancing imaging and sensing cap...

Huggingface_Bart_Cnn Summary:

Integrating functional materials with photonic and optoelectronic technologies has revolutionized medical diagnostics. Challenges such as material stability, reproducibility, scalability, and environmental sustainability remain critical barriers to their clinical translation. Breakthroughs such as g...

Huggingface_Pegasus_Pubmed Summary:

this review focuses on recent advances in the field of functional materials for medical diagnostics , such as photonics , plasmonic nanomaterials , organic semiconductors , and quantum dots , which are characterized by high sensitivity specificity and unique optical properties enabling realtime moni...

Huggingface_Led_Base Summary:

Abstract: ? . Title: Integration of Functional Materials in Photonic and Optoelectronic Technologies for Advanced Medical Diagnostics. _____ ? _____

_____ Abstract: Integrating functional materials with photonic ...

Huggingface_Bigbird_Pegasus Summary:

recent advances in materials science and technology have enabled a whole new world of materials for biomedical applications .<n> this article focuses on the field of diagnostic materials , which includes plasmonic semiconductors , quantum dots , and their electrochemical and optical properties , as ...

8. Tumour-infiltrating Lymphocytes and Radiation Therapy in Rectal Cancer: Systematic Review and

Source: PubMed

Authors: J Klein, W T Tran, S Viswanathan, R Salgado, P Poortmans, M Machiels

Year: 2024

Tumour-infiltrating lymphocytes (TILs) represent a promising cancer biomarker. Different TILs, including CD8+, CD4+, CD3+, and FOXP3+, have been associated with clinical outcomes. However, data are lacking regarding the value of TILs for patients receiving radiation therapy (RT). We conducted a systemic review and meta-analysis of available data evaluating TILs for patients receiving curative-intent therapy including RT.

Huggingface_Flan_T5 Summary:

Tumour-infiltrating lymphocytes and radiation therapy in rectal cancer: systemic review and meta-analysis: a systematic review of available data. [PubMed]NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/3.0/>). J Klein, W T Tran, S Viswanathan, R Salgado, P Poortmans, M Machiels.

Huggingface_Pegasus_Xsum Summary:

A systematic review and meta-analysis has been conducted to examine the value of infiltrating lymphocytes (s s) in radiation therapy for patients with rectal cancer, with a focus on the use of these biomarkers in the treatment of curative-intent cancer as well as in patients who are not candidates f...

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Huggingface_Pegasus_Pubmed Summary:

we report the discovery of a novel phosphatidylinositol 3-kinase (pi3k) <n> target gene , which was identified as a novel tumor suppressor in an orthotopic <n> mouse model of human gastric cancer . <n> the gene encodes a serine / threonine kinase that is <n> highly expressed on the surface of gast...

Huggingface_Led_Base Summary:

Abstract: ? | Title: Tumour-infiltrating Lymphocytes and Radiation Therapy in Rectal Cancer: Systematic Review and Meta-analysis. * () View . Type: Translational Tumor-activated lymphocytes (TILs) represent a promising cancer biomarker. Different TILs, including CD8+, CD4+, CD3+, ...

Huggingface_Bigbird_Pegasus Summary:

we present a genome - wide association study (gwas) that identifies snps in t - cell receptor (tcr) genes that are essential for normal t cell development and function .<n> we identify at least two new snps (atpase i and ii) in the cdkn2b / cdkn1a / rhoa - linked gene cluster of the human immu...

9. Microbiome characterization of patients with Crohn disease and the use of fecal microbiota trans

Source: PubMed

Authors: Shiju Chen, Daya Zhang, Da Li, Fan Zeng, Chen Chen, Feihu Bai

Year: 2025

Inflammatory bowel disease is a chronic inflammatory condition predominantly affecting the intestines, encompassing both ulcerative colitis and Crohn disease (CD). As one of the most common gastrointestinal disorders, CD's pathogenesis is closely linked with the intestinal microbiota. Recently, fecal microbiota transplantation (FMT) has gained attention as a potential treatment for CD, with the effective reestablishment of intestinal microecology considered a crucial mechanism of FMT therapy. Th...

Huggingface_Flan_T5 Summary:

Microbiome characterization of patients with Crohn disease and the use of fecal microbiota transplantation: a review of population-based cohort studies and the role of "beneficial" and "pathogenic" bacteria in the pathogenesis of Crohn's disease, inflammatory bowel disease, ulcerative colitis, and c...

Huggingface_Pegasus_Xsum Summary:

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Huggingface_Pegasus_Pubmed Summary:

backgroundfecal microbiota transplantation (fct) is an emerging treatment for inflammatory bowel disease (ibd) , ulcerative colitis (uc) and crohn 's disease . <n> the purpose of this article is to summarize the findings of recent studies on the fecal microbiota of patients with ibd and uc.met...

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Diagnostics & Detection

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Year: 2025

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Source: PubMed

Authors: Ping Zhang, Yuan Wang, Jianqi Bai, Jingru Zhang, Shimin Zhang, Xiaofei Guo, Jiawen Zhan, Liguu Zhu

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Huggingface_Pegasus_Pubmed Summary:

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Huggingface_Bigbird_Pegasus Summary:

advances in cell biology have been hampered by a lack of appropriate biomarkers for traumatic brain injury (tbi) .<n>

this study aimed to elucidate the biochemical and cellular mechanisms that underlie the transition from resting to sustained loading of the brain , which is the hallmark of tb . du...

3. A review of cutting-edge biomarkers for diagnosing coronary artery disease.

Source: PubMed

Authors: Pouria Azami, Sahand Mohammadzadeh, Soroush Seirafi, Iman Razeghian-Jahromi

Year: 2025

Chronic coronary artery disease (CAD) remains a significant global healthcare burden. Current risk assessment methods have notable limitations in early detection and risk stratification. Hence, there is an urgent need for innovative biomarkers that facilitate the premature CAD diagnosis, ultimately leading to reduction in associated morbidity and mortality rates. This review comprehensively examines recent advances in emerging biomarkers for CAD detection. Our analysis delves into various aspect...

Huggingface_Flan_T5 Summary:

A review of cutting-edge biomarkers for diagnosing coronary artery disease: a review of recent advances in emerging biomarker for the diagnosis of chronic coronary heart disease (CHD) and its risk stratification potentials: A systematic review of current and future research directions and review of ...

Huggingface_Pegasus_Xsum Summary:

Key words: biomarkers, coronary artery disease, early detection, risk stratification, Source: Pour Azami, Sahand Mohammadzadeh, Soroush Razeghian-Jahromi, Iman Razeghia, et al., Journal of the American College of Cardiology, 15(2), pages 1-17,doi:10.1542/ JAMA/Archives/Genetics, accessed on website:...

Huggingface_Mistral Summary:

Title: A review of cutting-edge biomarkers for diagnosing coronary artery disease.

Abstract: Chronic coronary artery disease (CAD) remains a significant global healthcare burden. Current risk assessment methods have notable limitations in early detection and risk stratification. Hen...

Huggingface_Bart_Cnn Summary:

Review comprehensively examines recent advances in emerging biomarkers for CAD detection. Tenascin-C, IL-37, PTX3, transthyretin, soluble interleukin-6 receptor ?, and miR-15a are identified as having high diagnostic potential for chronic CAD that indeed showcase promising performance metrics. Futur...

Huggingface_Pegasus_Pubmed Summary:

background : coronary artery disease (cad) is a major cause of morbidity and mortality worldwide . <n> although risk assessment methods have been developed , they remain limited in their ability to provide an early diagnosis of cad and to identify patients at high risk of developing the disease.me...

Huggingface_Led_Base Summary:

Title: A review of cutting-edge biomarkers for diagnosing coronary artery disease. ? _____
Author: Pouria Azami, Sahand Mohammadzadeh, Soroush Seirafi, and Iman Razeghian-Jahromi, et al. (2017)Abstract: Chronic cardiovascular disease (CAD) ...

Huggingface_Bigbird_Pegasus Summary:

advances in early diagnosis of coronary artery disease (cad) are hampered by a lack of sensitive and specific biomarkers .<n> this review examines the need for biomarkers , with a particular emphasis on transthyretin and

interleukin-6 , to improve cad diagnosis in the current environment of very l...

4. Ultrasensitive Detection of Circulating Plasma Cells Using Surface-Enhanced Raman Spectroscopy

Source: PubMed

Authors: Dechun Zhang, Xianling Chen, Jia Lin, Shiyang Jiang, Min Fan, Nenrong Liu, Zufang Huang, Jing Wang

Year: 2025

Multiple myeloma is a hematologic malignancy characterized by the proliferation of abnormal plasma cells in the bone marrow. Despite therapeutic advancements, there remains a critical need for reliable, noninvasive methods to monitor multiple myeloma. Circulating plasma cells (CPCs) in peripheral blood are robust and independent prognostic markers, but their detection is challenging due to their low abundance. Next-generation flow cytometry is commonly used for CPC detection but is not performed...

Huggingface_Flan_T5 Summary:

Ultrasensitive Detection of Circulating Plasma Cells Using Surface-Enhanced Raman Spectroscopy and Machine Learning for Multiple Myeloma Monitoring: A Cost-Effective, Rapid Assay Using Gold-Deposited Magnetic Nanoparticles and Plasmonic Nanoparticles Functionalized with Anti-CD138 and anti-CD38 Anti...

Huggingface_Pegasus_Xsum Summary:

A novel method for detecting circulating plasma cells in peripheral blood has been developed in the journal Nanoscale Science and Technology. The developed assay is highly sensitive and specific, capable of detecting as few as one or two cells, demonstrating excellent performance in differentiating ...

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Huggingface_Bart_Cnn Summary:

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Huggingface_Pegasus_Pubmed Summary:

background : multiple myeloma (mm) is a malignant neoplasm of the plasma cells . <n> it is characterized by the proliferation of abnormal plasma cells in the bone marrow and peripheral blood . because of their low abundance , flow cytometry is commonly used for their detection , but it is not perf...

Huggingface_Led_Base Summary:

Source: PubMed Abstract ? d Title: Ultrasensitive Detection of Circulating Plasma Cells Using Surface-Enhanced Raman Spectroscopy and Machine Learning for Multiple Myeloma Monitoring. * () Abstract: Multiple myeloma is a hematologic malignancy characterized by the proliferation of abn...

Huggingface_Bigbird_Pegasus Summary:

technological advances in the monitoring of multiple myeloma are numerous , but progress is slow . here , we review the progress being made in the detection , monitoring and prognostication of this incurable disease .<n> the need for more

sensitive and specific methods is evident but only a few have...

5. Turning Waste into Treasure: Functionalized Biomass-Derived Carbon Dots for Superselective Visualization and Eradication of Gram-Positive Bacteria

Source: PubMed

Authors: Ke-Fei Xu, Zihao Wang, Macheng Cui, Yuhan Jiang, Chengcheng Li, Zi-Xi Wang, Ling-Yi Li, Chenyang Jia, Lijie Zhang

Year: 2025

Gram-positive bacteria pose significant threats to human health, necessitating the development of targeted bacterial detection and eradication strategies. Nevertheless, current approaches often suffer from poor targeting specificity. Herein, the study utilizes purple rice lexivium to synthesize biomass carbon dots (termed BCDs) with wheat germ agglutinin-like residues for precisely targeting Gram-positive bacteria. Subsequently, fluorescein isothiocyanate (FITC) molecules are grafted onto BCDs t...

Huggingface_Flan_T5 Summary:

Turning Waste into Treasure: Functionalized Biomass-Derived Carbon Dots for Superselective Visualization and Eradication of Gram-Positive Bacteria. (<http://dx.doi.org/10.1016/j.sci.2015.09.018>.) - Authors: Ke-Fei Xu, Zihao Wang, Macheng Cui, Yuhan Jiang, Chengcheng Li, Zi-Xi Wang, Ling-Yi Li, Chenya...

Huggingface_Pegasus_Xsum Summary:

In this study, carbon dots (CDFs) produced from waste rice have the potential to be used to eradicate Gram-positive bacteria and to generate photodynamic therapy for the treatment of human health threats and infectious diseases such as Clostridium difficile (C. difficile) and MRSA (Methicillin-resis...

Huggingface_Mistral Summary:

Title: Turning Waste into Treasure: Functionalized Biomass-Derived Carbon Dots for Superselective Visualization and Eradication of Gram-Positive Bacteria.

Abstract: Gram-positive bacteria pose significant threats to human health, necessitating the development of targeted bacterial d...

Huggingface_Bart_Cnn Summary:

Study: Functionalized Biomass-Derived Carbon Dots for Superselective Visualization and Eradication of Gram-Positive Bacteria. Gram-positive bacteria pose significant threats to human health, necessitating the development of targeted bacterial detection and eradication strategies. Study: Rice lixiviu...

Huggingface_Pegasus_Pubmed Summary:

we report the discovery of a novel serine / threonine kinase (srk) , erk1 , in the endoplasmic reticulum (er) of drosophila melanogaster . <n> this enzyme functions as an er - specific transcription factor and is required for the establishment of er stress responses , which are essential for the...

Huggingface_Led_Base Summary:

Abstract: ? d Title: Turning Waste into Treasure: Functionalized Biomass-Derived Carbon Dots for Superselective Visualization and Eradication of Gram-Positive Bacteria. Translationalized BCDs (termed CDPs) with wheat germ agglutinin-like residues for precisely targeting Gram-positive bacteria....

Huggingface_Bigbird_Pegasus Summary:

although dna methylation has been recognized as a post - translational modification on chromatin , its precise role in regulating gene expression remains unclear .<n> here , we show that the small rna ubiquitin ligase ubc12 interacts with

ras homology 3 (rd3) to initiate histone methylation of his...

6. Unraveling Mechanisms of Genetic Risks in Metabolic Dysfunction-Associated Steatotic Liver Di

Source: PubMed

Authors: Xiang Zhang, Kyong-Mi Chang, Jun Yu, Rohit Loomba

Year: 2025

Metabolic dysfunction-associated steatotic liver disease (MASLD) is a growing global health problem, affecting ?1?billion people. This condition is well established to have a heritable component with strong familial clustering. With the extraordinary breakthroughs in genetic research techniques coupled with their application to large-scale biobanks, the field of genetics in MASLD has expanded rapidly. In this review, we summarize evidence regarding genetic predisposition to MASLD drawn from fami...

Huggingface_Flan_T5 Summary:

Unraveling Mechanisms of Genetic Risks in Metabolic Dysfunction-Associated Steatosis Liver Diseases: A Pathway to Precision Medicine. Xiang Zhang, Kyong-Mi Chang, Jun Yu, Rohit Loomba, et al. "Unravelling mechanisms of genetic risks in metabolic dysfunction-associated steatotic liver disease: a path...

Huggingface_Pegasus_Xsum Summary:

This article is published in Plos One, a journal of the American Association for the Advancement of Science (AAAS), on behalf of Xiang Chang, Jun Yu, and Rohit Loomba, all of the University of California, San Francisco (UCSD) School of Medicine, and their co-authors. Cite this article as "Unraveling...

Huggingface_Mistral Summary:

Title: Unraveling Mechanisms of Genetic Risks in Metabolic Dysfunction-Associated Steatotic Liver Diseases: A Pathway to Precision Medicine.

Abstract: Metabolic dysfunction-associated steatotic liver disease (MASLD) is a growing global health problem, affecting ?1?billion people. Th...

Huggingface_Bart_Cnn Summary:

Metabolic dysfunction-associated steatotic liver disease (MASLD) is a growing global health problem. This condition is well established to have a heritable component with strong familial clustering. We highlight the interplay between these genetic variants and their connections with metabolic factor...

Huggingface_Pegasus_Pubmed Summary:

we summarize evidence regarding genetic predisposition and we highlight the interplay between genetic variations and diverse pathogenic mechanisms . <n> we focus on the gut microbiota and its role in health and disease , which is driven by genetic factors and their interaction with environmental ris...

Huggingface_Led_Base Summary:

Title: Genetics in MASLD: A Pathway to Precision Medicine. ? I Title: Unraveling Mechanisms of Genetic Risks in Metabolic Dysfunction-Associated Steatotic Liver Diseases: APathway to precision Medicine. (2017)

_____...

Huggingface_Bigbird_Pegasus Summary:

advances in understanding the genetic basis of complex diseases have been slow . here , we summarize the progress being made in understanding genetic predisposition to , and consequences of , conditions ranging from liver disease to

autism . we find that , as our genome grows , so too does the compl...

7. Lipidomics of Huntington's Disease: A Comprehensive Review of Current Status and Future Directions

Source: PubMed

Authors: Ali Yilmaz, Sumeyya Akyol, Nadia Ashrafi, Nazia Saiyed, Onur Turkoglu, Stewart F Graham

Year: 2025

Huntington's disease (HD) is a multifaceted neurological disorder characterized by the progressive deterioration of motor, cognitive, and psychiatric functions. Despite a limited understanding of its pathogenesis, research has implicated abnormal trinucleotide cytosine-adenine-guanine CAG repeat expansion in the huntingtin gene (HTT) as a critical factor. The development of innovative strategies is imperative for the early detection of predictive biomarkers, enabling timely intervention and mitigation.

Huggingface_Flan_T5 Summary:

Lipidomics of Huntington's Disease: A Comprehensive Review of Current Status and Future Directions. [Published online ahead of print] Available at: http://www.ncbi.nlm.nih.gov/pubmed/2527891/pubmed_abstract.aspx?pub_id=217&page=1PubMed: <https://doi.org/10.1007/s11285-009-0038-008>

Huggingface_Pegasus_Xsum Summary:

The aim of this review is to provide a comprehensive review of the current status and future directions of Huntington's diseaseomics, with a particular focus on the development and use of novel biomarkers for the early detection of the disease and its potential treatment in patients with HD and other...

Huggingface_Mistral Summary:

Title: Lipidomics of Huntington's Disease: A Comprehensive Review of Current Status and Future Directions.

Abstract: Huntington's disease (HD) is a multifaceted neurological disorder characterized by the progressive deterioration of motor, cognitive, and psychiatric functions. Despite...

Huggingface_Bart_Cnn Summary:

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Huggingface_Pegasus_Pubmed Summary:

key clinical messagealthough rare , neurodegenerative disorders such as alzheimer s disease (ad) are a major public health problem worldwide . <n> understanding the genetic basis of ad is crucial for developing effective therapies . in this study , we used whole - genome transcriptome sequencing t...

Huggingface_Led_Base Summary:

Title: Lipidomics of Huntington's Disease: A Comprehensive Review of Current Status and Future Directions. PubMed Abstract: Lipidomics of Parkinson's Disease. PubMed Type: Intra-Lipidomics. ? View Abstract: Huntington's disease (HD) is a multifaceted neurological disorder characterized by the...

Huggingface_Bigbird_Pegasus Summary:

understanding the biology of neurodegenerative disorders has been hampered by a lack of methods that allow for the assessment of pathological changes in the course of disease .<n> the development of highly sensitive and specific

methods for the study of neuritic diseases has benefited greatly from a...

8. Integration of Functional Materials in Photonic and Optoelectronic Technologies for Advanced Medical Diagnostics

Source: PubMed

Authors: Naveen Thanjavur, Laxmi Bugude, Young-Joon Kim

Year: 2025

Integrating functional materials with photonic and optoelectronic technologies has revolutionized medical diagnostics, enhancing imaging and sensing capabilities. This review provides a comprehensive overview of recent innovations in functional materials, such as quantum dots, perovskites, plasmonic nanomaterials, and organic semiconductors, which have been instrumental in the development of diagnostic devices characterized by high sensitivity, specificity, and resolution. Their unique optical p...

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Integrating Functional Materials in Photonic and Optoelectronic Technologies for Advanced Medical Diagnostics: A Comprehensive Review of Recent Innovations in Functional Materials, such as Quantum Dots, Perovskites, Plasmonic Nanomaterials, and Organic Semiconductor Devices, which have been instrumental in the development of diagnostic devices characterized by high sensitivity, specificity, and resolution.

Huggingface_Pegasus_Xsum Summary:

Functional materials are playing a key role in the development and manufacture of next-generation medical diagnostics devices, which have the potential to transform the way we diagnose and treat diseases such as cancer, cardiovascular disease, diabetes, and neurodegenerative disorders. This article ...

Huggingface_Mistral Summary:

Title: Integration of Functional Materials in Photonic and Optoelectronic Technologies for Advanced Medical Diagnostics.

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Huggingface_Bart_Cnn Summary:

Integrating functional materials with photonic and optoelectronic technologies has revolutionized medical diagnostics. Challenges such as material stability, reproducibility, scalability, and environmental sustainability remain critical barriers to their clinical translation. Breakthroughs such as graphene-based photonic devices and quantum dot-based sensors have shown promise in overcoming these challenges.

Huggingface_Pegasus_Pubmed Summary:

this review focuses on recent advances in the field of functional materials for medical diagnostics, such as photonics, plasmonic nanomaterials, organic semiconductors, and quantum dots, which are characterized by high sensitivity, specificity, and unique optical properties enabling real-time monitoring and diagnosis.

Huggingface_Led_Base Summary:

Abstract: ? . Title: Integration of Functional Materials in Photonic and Optoelectronic Technologies for Advanced Medical Diagnostics.

Abstract: Integrating functional materials with photonic ...

Huggingface_Bigbird_Pegasus Summary:

recent advances in materials science and technology have enabled a whole new world of materials for biomedical applications. this article focuses on the field of diagnostic materials, which includes plasmonic semiconductors, quantum dot-based sensors, and graphene-based photonic devices.

quantum dots , and their electrochemical and optical properties , as ...

9. Predictive model of intrahepatic collateral vessels among hepatic veins in patients with liver tumors

Source: PubMed

Authors: Fabio Procopio, Jacopo Galvanin, Giulia Mauri, Simone Famularo, Vittorio Pedicini, Flavio Milana, Guido Costa, Guido

Year: 2025

Communicating vessels among hepatic veins in patients with tumors invading/compressing hepatic veins at their caval confluence facilitate new surgical solutions. Although their recognition by intraoperative ultrasound has been described, the possibility of preoperative detection still remains uncertain. We aimed to develop a model to predict their presence before surgery.

Huggingface_Flan_T5 Summary:

Predictive Model of Intrahepatic Collateral Veins Among Hepatic Venouses in Patients with Liver Tumors Associated with the Caval Confluence: A Prospective Model for Preoperative Detection and Preoperative Management of Tumor Invading/Compressing Hepatitis C Virus-Infected Liver Diseases (PubMed Author...

Huggingface_Pegasus_Xsum Summary:

The aim of this study was to develop a model to predict before surgery the presence of vessels at the caval confluence, which could be used to identify patients with liver disease who are at risk of complications during surgery and to prevent them from undergoing further treatment after the procedur...

Huggingface_Mistral Summary:

Title: Predictive model of intrahepatic collateral vessels among hepatic veins in patients with liver tumors involving the caval confluence.

Abstract: Communicating vessels among hepatic veins in patients with tumors invading/compressing hepatic veins at their caval confluence facil...

Huggingface_Bart_Cnn Summary:

Communicating vessels among hepatic veins in patients with tumors invading/compressing hepatic. veins at their caval confluence. Although their recognition by intraoperative ultrasound has been described, the possibility of preoperative detection still remains uncertain. We aimed to develop a model ...

Huggingface_Pegasus_Pubmed Summary:

we describe a model of intrahepatic collateral vessels among hepatic tumors involving caval vessels . <n> the presence of caval blood vessels was detected by intraoperative ultrasound in a patient with hepatic metastases and multiple caval veins in the liver . to our knowledge , this is the first re...

Huggingface_Led_Base Summary:

Abstract: ? d Title: Predictive model of intrahepatic collateral vessels among hepatic veins in patients with liver tumors involving the caval confluence. ? ? ? D Abstract: Communicating vessels among hepatitisatic veinsIn patients with tumors invading/compressing hepatic vein at thei...

Huggingface_Bigbird_Pegasus Summary:

although arteriovenous shunts (avs) are widely used in neurosurgical procedures , the use of avs in hepatobiliary surgery remains controversial .<n> here , we review the literature and present our recent experience with avs during hepatic tumor resection . as an initial step toward developing a su...

10. A review of cutting-edge biomarkers for diagnosing coronary artery disease.

Source: PubMed

Authors: Pouria Azami, Sahand Mohammadzadeh, Soroush Seirafi, Iman Razeghian-Jahromi

Year: 2025

Chronic coronary artery disease (CAD) remains a significant global healthcare burden. Current risk assessment methods have notable limitations in early detection and risk stratification. Hence, there is an urgent need for innovative biomarkers that facilitate the premature CAD diagnosis, ultimately leading to reduction in associated morbidity and mortality rates. This review comprehensively examines recent advances in emerging biomarkers for CAD detection. Our analysis delves into various aspect...

Huggingface_Flan_T5 Summary:

A review of cutting-edge biomarkers for diagnosing coronary artery disease: a review of recent advances in emerging biomarker for the diagnosis of chronic coronary heart disease (CHD) and its risk stratification potentials: A systematic review of current and future research directions and review of ...

Huggingface_Pegasus_Xsum Summary:

Key words: biomarkers, coronary artery disease, early detection, risk stratification, Source: Pour Azami, Sahand Mohammadzadeh, Soroush Razeghian-Jahromi, Iman Razeghia, et al., Journal of the American College of Cardiology, 15(2), pages 1-17,doi:10.1542/ JAMA/Archives/Genetics, accessed on website:...

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_____ Author: Pouria Azami, Sahand Mohammadzadeh, Soroush Seirafi, and Iman Razeghian-Jahromi, et al. (2017)Abstract: Chronic cardiovascular disease (CAD) ...

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Medical Devices

1. Ultrasensitive Detection of Circulating Plasma Cells Using Surface-Enhanced Raman Spectroscopy and Machine Learning for Multiple Myeloma Monitoring: A Cost-Effective, Rapid Assay Using Gold-Deposited Magnetic Nanoparticles and Plasmonic Nanoparticles Functionalized with Anti-CD138 and anti-CD38 Antibodies

Source: PubMed

Authors: Dechun Zhang, Xianling Chen, Jia Lin, Shiyang Jiang, Min Fan, Nenrong Liu, Zufang Huang, Jing Wang

Year: 2025

Multiple myeloma is a hematologic malignancy characterized by the proliferation of abnormal plasma cells in the bone marrow. Despite therapeutic advancements, there remains a critical need for reliable, noninvasive methods to monitor multiple myeloma. Circulating plasma cells (CPCs) in peripheral blood are robust and independent prognostic markers, but their detection is challenging due to their low abundance. Next-generation flow cytometry is commonly used for CPC detection but is not performed...

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Huggingface_Pegasus_Xsum Summary:

Error: 500 Server Error: Internal Server Error for url: <https://api-inference.huggingface.co/models/google/pegasus-xsum> | Response: {"error": "Model too busy, unable to get response in less than 60 second(s)"}

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3. Evaluating the performance of a new ergonomic laparoscopic needle holder for intracorporeal suturing

Source: PubMed

Authors: Arturo Minor-Martínez, Ricardo Manuel Ordorica-Flores, Juan Ramón Mota-Carmona, Iván Tlacuélle Franco-González

Year: 2025

In laparoscopy, the absence of ergonomics in the instruments affects the performance and efficiency of the surgeon, increasing the likelihood of developing musculoskeletal injuries. This article presents the development of a laparoscopic needle holder with an ergonomic handle and the experience in its use with surgeons in the intracorporeal suturing task. The handle of the laparoscopic needle holder consists of a semi-spherical piece that easily adapts to the palm of the surgeon's hand and improves...

Huggingface_Flan_T5 Summary:

Evaluating the performance of a new ergonomic laparoscopic needle holder for intracorporeal suturing: a pilot study with pediatric surgery surgeons and residents in a laparoscopy simulator. (Journal of the American College of Surgeons, Vol. 62, No. 1, pp. 87-94) [Review] The laparoscopic needle holder...

Huggingface_Pegasus_Xsum Summary:

A new ergonomic laparoscopic needle holder for intracorporeal suturing has been developed and tested in a laparoscopy simulator, with the aim of improving the performance and ergonomics of surgeons in this surgical procedure, as well as reducing the risk of musculoskeletal injuries to the surgeon and...

Huggingface_Mistral Summary:

Title: Evaluating the performance of a new ergonomic laparoscopic needle holder for intracorporeal suturing. Abstract: In laparoscopy, the absence of ergonomics in the instruments affects the performance and efficiency of the surgeon, increasing the likelihood of developing musculoskeletal injuries....

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Huggingface_Pegasus_Pubmed Summary:

introduction : article presents the development laparoscopic needle holder with an ergonomic handle and experience with surgeons in intracorporeal suturing task.material and methods : two pediatric surgeons participated in this study .
<n> data from both instruments were captured and the performance...

Huggingface_Led_Base Summary:

Title: Evaluating the performance of a new ergonomic laparoscopic needle holder for intracorporeal suturing. Abstract: In laparoscopy, the absence of ergonomics in the instruments affects the performance and efficiency of the surgeon, increasing the likelihood of developing musculoskeletal injuries....

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4. Integration of Functional Materials in Photonic and Optoelectronic Technologies for Advanced Medical Applications

Source: PubMed

Authors: Naveen Thanjavur, Laxmi Bugude, Young-Joon Kim

Year: 2025

Integrating functional materials with photonic and optoelectronic technologies has revolutionized medical diagnostics, enhancing imaging and sensing capabilities. This review provides a comprehensive overview of recent innovations in functional materials, such as quantum dots, perovskites, plasmonic nanomaterials, and organic semiconductors, which have been instrumental in the development of diagnostic devices characterized by high sensitivity, specificity, and resolution. Their unique optical p...

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Huggingface_Led_Base Summary:

Abstract: ? . Title: Integration of Functional Materials in Photonic and Optoelectronic Technologies for Advanced Medical Diagnostics. _____ ? _____

_____ Abstract: Integrating functional materials with photonic ...

Huggingface_Bigbird_Pegasus Summary:

recent advances in materials science and technology have enabled a whole new world of materials for biomedical applications. <n> this article focuses on the field of diagnostic materials, which includes plasmonic semiconductors, quantum dots, and their electrochemical and optical properties, as ...

Other Research

1. Female genital mutilation among children in Ethiopia: A time-to-event analysis of age at circumcision

Source: PubMed

Authors: Michael Arthur Ofori, Daniel Biftu Bekalo, David Kwamena Mensah, Niharika Jha

Year: 2025

Female Genital Mutilation (FGM) has become a global health concern. It is a deeply entrenched harmful practice involving partial or total removal of the external female genitalia for non-medical reasons. To inform effective policymaking and raise awareness about FGM's health risks, understanding socioeconomic and demographic factors influencing the timing of girls' circumcision is crucial. This study employed semi-parametric survival models to examine the association between residential status a...

Huggingface_Flan_T5 Summary:

Female genital mutilation among children in Ethiopia: A time-to-event analysis of age at circumcision: a population-based study of the 2016 Ethiopian Demographic and Health Survey (EDHS): A case-control study of female circumcision among children aged 0 to 24 years. [Ethnicity, religion, mother's ed...

Huggingface_Pegasus_Xsum Summary:

Title: Female genital mutilation among children in Ethiopia: A time-to-event analysis of age at circumcision, International Journal of Health and Social Care (IJHSC), Forthcoming, Available at: <https://doi.org/10.2556/ijhssc.2017.07.023>.<n> The analysis revealed residential status (urban vs rural, HR =...

Huggingface_Mistral Summary:

Title: Female genital mutilation among children in Ethiopia: A time-to-event analysis of age at circumcision.

Abstract: Female Genital Mutilation (FGM) has become a global health concern. It is a deeply entrenched harmful practice involving partial or total removal of the external f...

Huggingface_Bart_Cnn Summary:

Female Genital Mutilation (FGM) has become a global health concern. It is a deeply entrenched harmful practice involving partial or total removal of the external female genitalia for non-medical reasons. To inform effective policymaking and raise awareness about FGM's health risks, understanding soc...

Huggingface_Pegasus_Pubmed Summary:

backgroundthe aim of this study was to determine the association between residential status of girls and their semi-circumcision.methodsthis was a cross - sectional study conducted in 2016 among girls aged 15 years and older in saudi arabia . <n> data used in this report was residential and that was...

Huggingface_Led_Base Summary:

Title: Female genital mutilation among children in Ethiopia: A time-to-event analysis of age at circumcision. Source: PubMed Abstract | CrossRef Full Text | Google Scholar | Author: Michael Arthur Ofori, Daniel Biftu Bekalo, David Kwamena Mensah, Niharika Jha, et al. ? _____...

Huggingface_Bigbird_Pegasus Summary:

the world health organization (who) recommends that all girls be circumcised at the age of 12 years .<n> however , in many parts of the world , mutilation is performed for religious reasons . in this study , we examined the impact of non - religious reasons on girls ' initiation of circumcision , ...

2. Exploring Health Educational Interventions for Children With Congenital Heart Disease: Scoping

Source: PubMed

Authors: Neda Barbazi, Ji Youn Shin, Gurumurthy Hiremath, Carlye Anne Lauff

Year: 2025

Congenital heart disease (CHD) is the most common birth defect, affecting 40,000 births annually in the United States. Despite advances in medical care, CHD is often a chronic condition requiring continuous management and education. Effective care management depends on children's understanding of their condition. This highlights the need for targeted health educational interventions to enhance health literacy among children with CHD.

Huggingface_Flan_T5 Summary:

Exploring Health Educational Interventions for Children With Congenital Heart Disease: Scoping Review. [PubMed]NC-ND license: Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License: CC-BY-NC 3.0]. Authors: Neda Barbazi, Ji Youn Shin, Gurumurthy Hiremath, Carlye Anne Lauff...

Huggingface_Pegasus_Xsum Summary:

The purpose of this review is to explore health educational interventions for children with Congenital Heart Disease (CHD) and to highlight the need for targeted educational interventions to enhance health literacy among children with CHD., published online by the American College of Cardiology (ACC...

Huggingface_Mistral Summary:

Title: Exploring Health Educational Interventions for Children With Congenital Heart Disease: Scoping Review.

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Huggingface_Pegasus_Pubmed Summary:

we report the case of a 4-year - old boy who was born with a ruptured abdominal aortic aneurysm (oa) . at birth , <n> the patient presented with bilateral abdominal pain . <n> computed tomography (ct) revealed a large right abdominal mass , and magnetic resonance imaging (mri) confirmed the oa...

Huggingface_Led_Base Summary:

Abstract: ? . Title: Exploring Health Educational Interventions for Children With Congenital Heart Disease: Scoping Review. ? (1) d , Abstract:Congenital heart disease (CHD) is the most common birth defect, affecting 40,000 births annually in the United States. Despite advances in medi...

Huggingface_Bigbird_Pegasus Summary:

key clinical messagea rare cause of parkinson 's disease is the underdosing of dopamine in dopaminergic neurons in the substantia nigra .<n> dopamine can be reuptake in the basal ganglia , but not in the cortex . here , we show that the axon / exo - inositol / glucagon (a / e / g) pathway plays a ...

3. A New Deep Learning-Based Method for Automated Identification of Thoracic Lymph Node Station

Source: PubMed

Authors: yvind Ervik, Mia R?dde, Erlend Fagertun Hofstad, Ingrid Tveten, Thomas Lang?, H?kon O Leira, Tore Amundsen, Ha

Year: 2025

Endobronchial ultrasound-guided transbronchial needle aspiration (EBUS-TBNA) is a cornerstone in minimally invasive thoracic lymph node sampling. In lung cancer staging, precise assessment of lymph node position is crucial for clinical decision-making. This study aimed to demonstrate a new deep learning method to classify thoracic lymph nodes based on their anatomical location using EBUS images. Bronchoscopists labeled lymph node stations in real-time according to the Mountain Dressler nomenclat...

Huggingface_Flan_T5 Summary:

A New Deep Learning-Based Method for Automated Identification of Thoracic Lymph Node Stations in Endobronchial Ultrasound (EBUS): A Proof-of-Concept Study. yvind Ervik, Mia Rdde, Erlend Fagertun Hofstad, Ingrid Tveten, Thomas Lang, Hkon O Leira, Tore Amundsen, Hanne Sorger

Huggingface_Pegasus_Xsum Summary:

A new deep learning-based method for automated identification of Thoracic Lymph Node Stations in Endobronchial Ultrasound (EBUS) has been demonstrated in a proof-of-concept study published in the Journal of the American College of Surgeons (JACS) by researchers from the Norwegian University of Scien...

Huggingface_Mistral Summary:

Title: A New Deep Learning-Based Method for Automated Identification of Thoracic Lymph Node Stations in Endobronchial Ultrasound (EBUS): A Proof-of-Concept Study.

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Huggingface_Bart_Cnn Summary:

Study aimed to demonstrate a new deep learning method to classify thoracic lymph nodes. Bronchoscopists labeled lymph node stations in real-time according to the Mountain Dressler nomenclature. 28,134 EBUS images were acquired from 56 patients. Highest precision, sensitivity, and F1 score were obser...

Huggingface_Pegasus_Pubmed Summary:

a new learning method for lymph node classification is demonstrated . <n> first , real lymph node stations were labeled with neural network images . then , image processing was performed on image sets , resulting in neural network maps . from these maps , <n> lymph nodes were classified according to...

Huggingface_Led_Base Summary:

Abstract: ? I Title: A New Deep Learning-Based Method for Automated Identification of Thoracic Lymph Node Stations in Endobronchial Ultrasound (EBUS): A Proof-of-Concept Study. ? Type: Abstract Abstract: Efficiently Assessing the Orthopaedic Location of Leukemia Stations Using EBUS Images in...

Huggingface_Bigbird_Pegasus Summary:

advances in precision medicine have the potential to improve the quality of life of cancer patients by reducing the time and expense of lymph node staging .<n> this paper describes the development and validation of a new image - guided dissection technique that uses automated high - resolution , thr...

4. Investigating the relationship between social media use and the attitudes towards nutrition and b

Source: PubMed

Authors: Sogol Keyvanpour, Masoumeh Namazi, Fatemeh Hosseini, Narges Ebrahimi

Year: 2025

Social media are Internet-based services that allow participation in online communities and exchanges. Considering the high and increasing statistics of the use of social media all over the world and its impact on people's lives, the present study aimed to determine the relationship between social media and nutritional attitudes and body image shame among Iranian female students. This cross-sectional study was performed on 201 female student of Tehran University of Medical Sciences in Tehran, Ir...

Huggingface_Flan_T5 Summary:

social media usage had a significant and positive indirect effect on the indicator of body image shame of students (P .05). Therefore, the relationship between social media use and the attitudes towards nutrition and body image Shame among Iranian female students should be further investigated in o...

Huggingface_Pegasus_Xsum Summary:

The aim of this study was to investigate the relationship between social media use and the attitudes towards nutrition and body image shame among female students in Tehran University of Medical Sciences in Tehran, Iran, from May to December 2023, using a cross-sectional study method.; 2011;chap.: 1-...

Huggingface_Mistral Summary:

Title: Investigating the relationship between social media use and the attitudes towards nutrition and body image shame among Iranian female students: A cross-sectional study. Abstract: Social media are Internet-based services that allow participation in online communities and exchanges. Considering...

Huggingface_Bart_Cnn Summary:

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Huggingface_Pegasus_Pubmed Summary:

aim : to determine the relationship between social media use and attitudes towards nutrition and body image among female students.materials and methods : cross-sectional study was performed on 201 female students from student to 2023 using paper form . collection tools included demographic informati...

Huggingface_Led_Base Summary:

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Huggingface_Bigbird_Pegasus Summary:

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Clinical Trials

1. A review of cutting-edge biomarkers for diagnosing coronary artery disease.

Source: PubMed

Authors: Pouria Azami, Sahand Mohammadzadeh, Soroush Seirafi, Iman Razeghian-Jahromi

Year: 2025

Chronic coronary artery disease (CAD) remains a significant global healthcare burden. Current risk assessment methods have notable limitations in early detection and risk stratification. Hence, there is an urgent need for innovative biomarkers that facilitate the premature CAD diagnosis, ultimately leading to reduction in associated morbidity and mortality rates. This review comprehensively examines recent advances in emerging biomarkers for CAD detection. Our analysis delves into various aspect...

Huggingface_Flan_T5 Summary:

A review of cutting-edge biomarkers for diagnosing coronary artery disease: a review of recent advances in emerging biomarker for the diagnosis of chronic coronary heart disease (CHD) and its risk stratification potentials: A systematic review of current and future research directions and review of ...

Huggingface_Pegasus_Xsum Summary:

Key words: biomarkers, coronary artery disease, early detection, risk stratification, Source: Pour Azami, Sahand Mohammadzadeh, Soroush Razeghian-Jahromi, Iman Razeghia, et al., Journal of the American College of Cardiology, 15(2), pages 1-17,doi:10.1542/ JAMA/Archives/Genetics, accessed on website:...

Huggingface_Mistral Summary:

Title: A review of cutting-edge biomarkers for diagnosing coronary artery disease.

Abstract: Chronic coronary artery disease (CAD) remains a significant global healthcare burden. Current risk assessment methods have notable limitations in early detection and risk stratification. Hen...

Huggingface_Bart_Cnn Summary:

Review comprehensively examines recent advances in emerging biomarkers for CAD detection. Tenascin-C, IL-37, PTX3, transthyretin, soluble interleukin-6 receptor ?, and miR-15a are identified as having high diagnostic potential for chronic CAD that indeed showcase promising performance metrics. Futur...

Huggingface_Pegasus_Pubmed Summary:

background : coronary artery disease (cad) is a major cause of morbidity and mortality worldwide . <n> although risk assessment methods have been developed , they remain limited in their ability to provide an early diagnosis of cad and to identify patients at high risk of developing the disease.me...

Huggingface_Led_Base Summary:

Title: A review of cutting-edge biomarkers for diagnosing coronary artery disease. ? _____
_____ Author: Pouria Azami, Sahand Mohammadzadeh, Soroush Seirafi, and Iman Razeghian-Jahromi, et al. (2017)Abstract: Chronic cardiovascular disease (CAD) ...

Huggingface_Bigbird_Pegasus Summary:

advances in early diagnosis of coronary artery disease (cad) are hampered by a lack of sensitive and specific biomarkers .<n> this review examines the need for biomarkers , with a particular emphasis on transthyretin and interleukin-6 , to improve cad diagnosis in the current environment of very l...

2. Efficacy of a clinical decision support tool to promote guideline concordant evaluations in patients with high-risk microscopic hematuria: a cluster randomized quality improvement project

Source: PubMed

Authors: Richard S Matulewicz, Sarah Tsuruo, William C King, Arielle R Nagler, Zachary S Feuer, Adam Szerencsy, Danil V Matulewicz

Year: 2025

We aimed to determine whether implementation of clinical decision support (CDS) tool integrated into the electronic health record (EHR) of a multi-site academic medical center increased the proportion of patients with American Urological Association (AUA) "high risk" microscopic hematuria (MH) who receive guideline concordant evaluations.

Huggingface_Flan_T5 Summary:

Efficacy of a clinical decision support tool to promote guideline concordant evaluations in patients with high-risk microscopic hematuria: a cluster randomized quality improvement project: A multi-site academic medical center experience. Clinical Decision Support (CDS) is an electronic health record...

Huggingface_Pegasus_Xsum Summary:

Title: Efficacy of a clinical decision support tool to promote guideline concordant evaluations in patients with high-risk microscopic hematuria: a randomized quality improvement project., Journal of the American Urological Association, 15(2), 21-24.<n> The authors report that implementation of a CD...

Huggingface_Mistral Summary:

Title: Efficacy of a clinical decision support tool to promote guideline concordant evaluations in patients with high-risk microscopic hematuria: a cluster randomized quality improvement project.

Abstract: We aimed to determine whether implementation of clinical decision support (CD...

Huggingface_Bart_Cnn Summary:

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Huggingface_Pegasus_Pubmed Summary:

introduction : electronic health records (ehra) are becoming increasingly important in clinical practice . <n> however , their implementation in routine clinical practice has not yet been established . here <n> , we propose a concordant decision tool (cdr) to facilitate the implementation of hig...

Huggingface_Led_Base Summary:

Abstract: ? d Title: Efficacy of a clinical decision support tool to promote guideline concordant evaluations in patients with high-risk microscopic hematuria: a cluster randomized quality improvement project. ? o Abstract: We aimed to determine whether implementation of clinical decisio...

Huggingface_Bigbird_Pegasus Summary:

advances in microrna (mirna) biology are proceeding at a rapid rate with the discovery of new genes and new targets .<n> however , because mirna biology is still in its infancy , there is a great need for guidelines that provide a clear picture of how to use this new technology in clinical practic...

3. A review of cutting-edge biomarkers for diagnosing coronary artery disease.

Source: PubMed

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Public Health

1. Evaluation of an online Cardiometabolic and Weight Loss Program. A mixed methods study.

Source: PubMed

Authors: Dianna A Dandeneau, Silvia J Leon, Margherita Cameranesi, Mackenzie R Alexiuk, Jennifer Lp Protudjer, Clara J Boh

Year: 2025

Risk factors contributing to cardiovascular diseases (CVD) can be addressed through behavior modification, including changes in diet and physical activity. In 2021, The Wellness Institute (WI), located at Seven Oaks General Hospital, created a virtual cardiometabolic risk reduction program in response to COVID-19 pandemic public health restrictions, encompassing virtual health coaching and lifestyle education. The objective was to evaluate the acceptability, adherence, efficacy and engagement of...

Huggingface_Flan_T5 Summary:

Evaluation of an online Cardiometabolic and Weight Loss Program. A Mixed Methods Study - Seven Oaks General Hospital, COVID-19 Pandemic Public Health Restrictions, 2021, 93 participants, ages 58 11 years, weight 98 20.9 kg, p0.001) The Wellness Institute (WI) created a virtual cardiometabolic risk...

Huggingface_Pegasus_Xsum Summary:

The purpose of this study was to evaluate the acceptability and efficacy of an online cardiometabolic and weight loss program in response to CO-19 pandemic risk reduction restrictions, encompassing virtual health coaching and virtual health education restrictions, at Seven Oaks General Hospital, in ...

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Huggingface_Pegasus_Pubmed Summary:

abstractthe aim of this study was to evaluate the efficacy of an online coaching program for weight loss and to assess its acceptability . <n> participants (n = 93) were randomly assigned to the control group (no coaching) or the experimental group (four weekly coaching sessions , once a week ,...

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Huggingface_Bigbird_Pegasus Summary:

the prevalence of diabetes is increasing at an alarming rate in the developed world .<n> the reasons for this increase are not fully understood . over the last few decades ,<n> lifestyle modification programs have been developed with the primary objective to prevent or delay the development of obesi...

2. Microbiome characterization of patients with Crohn disease and the use of fecal microbiota trans

Source: PubMed

Authors: Shiju Chen, Daya Zhang, Da Li, Fan Zeng, Chen Chen, Feihu Bai

Year: 2025

Inflammatory bowel disease is a chronic inflammatory condition predominantly affecting the intestines, encompassing both ulcerative colitis and Crohn disease (CD). As one of the most common gastrointestinal disorders, CD's pathogenesis is closely linked with the intestinal microbiota. Recently, fecal microbiota transplantation (FMT) has gained attention as a potential treatment for CD, with the effective reestablishment of intestinal microecology considered a crucial mechanism of FMT therapy. Th...

Huggingface_Flan_T5 Summary:

Microbiome characterization of patients with Crohn disease and the use of fecal microbiota transplantation: a review of population-based cohort studies and the role of "beneficial" and "pathogenic" bacteria in the pathogenesis of Crohn's disease, inflammatory bowel disease, ulcerative colitis, and c...

Huggingface_Pegasus_Xsum Summary:

Microbiome characterization of patients with Crohn disease and the use of fecal microbiota transplantation: A review.<n> This article synthesizes the findings of population-based cohort studies to enhance our understanding of gut microbial characteristics in patients with CD, and it delves into the ...

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Huggingface_Pegasus_Pubmed Summary:

backgroundfecal microbiota transplantation (fct) is an emerging treatment for inflammatory bowel disease (ibd) , ulcerative colitis (uc) and crohn 's disease . <n> the purpose of this article is to summarize the findings of recent studies on the fecal microbiota of patients with ibd and uc.met...

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Source: PubMed

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Year: 2025

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Key Takeaways & Recommendations

Based on today's research findings:

- 1. Treatment Advances: - 9 new studies on treatments - Focus on personalized medicine and targeted therapies
- 2. Diagnostic Improvements: - 10 new diagnostic approaches - Emphasis on early detection and precision diagnostics
- 3. Technology Integration: - 4 new medical devices and technologies - Trend towards AI-enabled and smart medical devices
- 4. Public Health Implications: - 3 public health studies - Important findings for population health management

Recommended Actions:

- 1. Review promising treatments in clinical trials for potential fast-track approval
- 2. Evaluate new diagnostic tools for integration into healthcare systems
- 3. Consider pilot programs for innovative medical devices
- 4. Update public health guidelines based on new findings